

Agent Run Report — Healthcare Q&A Agent

A short walkthrough of what I built, how it thinks, and how it behaves when you actually run it. The goal was a small, honest agent that answers health questions by looking things up and summarizing what it finds — not a giant framework, and nothing that needs servers to run.

Live demo: <https://healthcare-agent-fo48sgeutc25uy2ogrpk6.streamlit.app/>

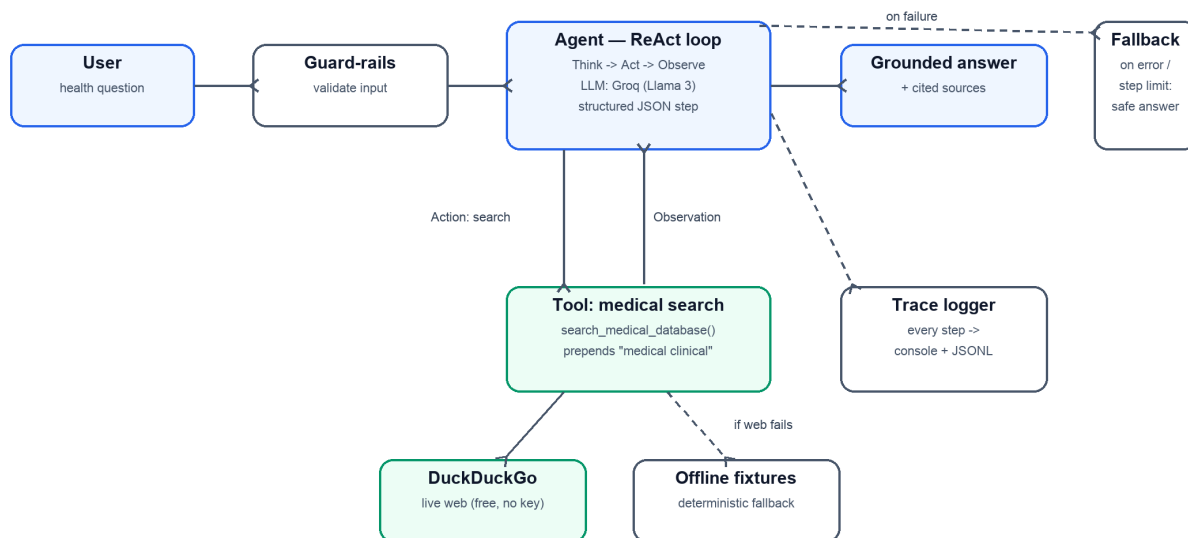
1. What it does, in one line

You ask a health question, the agent searches a medical knowledge base, reads the results, and writes back a short answer with the sources it used.

I picked the **healthcare Q&A** domain because it shows the whole agent story clearly: it has to decide *what* to look up, actually call a tool, read the results, and then ground its answer in them.

2. Architecture

Healthcare Q&A Agent — Architecture



The flow is a classic **ReAct loop** (Reason → Act → Observe), which mirrors how a person researches something:

- **Guard-rails** — sanity-check the question before spending any tokens.
- **Agent (the LLM)** — each turn returns one small JSON step: a *thought* plus an *action* (either "search" or "final answer").
- **Tool: medical search** — `search_medical_database()` uses DuckDuckGo (free, no key) as a stand-in for PubMed, steering queries toward medical sources.
- **Observation** — the top results (title, snippet, link) go back to the agent.

- The loop repeats until the agent is confident, then it writes a **grounded answer with cited sources**.

Two supporting pieces make it robust: a **trace logger** that records every step to the console and a JSONL file, and a **fallback** that returns a safe answer if the LLM errors or the step budget runs out (and the search tool falls back to bundled fixtures if the web is unavailable).

Why these choices

- **Free + zero infra:** Groq's free tier (Llama 3) for the LLM, DuckDuckGo for search.
- **Custom loop, not a framework:** the deliverable is the reasoning trace, so I kept the loop small (~40 lines) and readable instead of hiding it in LangChain.
- **Structured JSON every step:** validated with Pydantic for clean parsing.

3. The app in action

A simple Streamlit UI wraps the same agent. These are real runs (Groq + live DuckDuckGo search).

Example A — "What is the treatment for the common cold?"

The screenshot shows a Streamlit application titled "Healthcare Q&A Agent". On the left, a sidebar titled "How it works" lists four steps: 1. Think about the question, 2. Search a medical knowledge base (DuckDuckGo as a PubMed stand-in), 3. Observe the results, and 4. Answer, grounded in sources. Below this is a blue box with the text "Educational demo only — not medical advice." The main area shows the query "what is treatment for common cold ?" in a text input, followed by a red "Ask" button. Below the button, the "Answer" section states: "The common cold is typically managed by alleviating symptoms and preventing complications. Treatment may include over-the-counter medications for pain and fever, rest, and hydration. This is general information, not a substitute for professional medical advice." The "Sources" section lists three URLs: <https://www.cdc.gov/common-cold/treatment/index.html>, [https://www.jaci-global.org/article/S2772-8293\(24\)00114-0/fulltext](https://www.jaci-global.org/article/S2772-8293(24)00114-0/fulltext), and <https://www.uptodate.com/contents/the-common-cold-in-adults-treatment-and-prevention>. At the bottom, a "Reasoning trace" section shows "Steps taken: 3" and a "Thought: I need to find evidence-based recommendations for treating the common cold."

The agent answers and lists the sources it actually used.

The nice part is the trace below: the agent's first search returned weak results, so it noticed and searched again with a better query before answering — genuine reasoning, not a single canned lookup:

How it works

1. Think about the question
2. Search a medical knowledge base (DuckDuckGo as a PubMed stand-in)
3. Observe the results
4. Answer, grounded in sources

Educational demo only — not medical advice.

Deploy

Reasoning trace

Steps taken: 3

Thought: I need to find evidence-based recommendations for treating the common cold.

Action: `search_medical_database('treatment for the common cold')`

Observation: retrieved 3 results — Common Cold Treatment Clinic Near Me in Irving TX; How to Prevent and Treat the Common Cold | Taylor Wolfram; Common cold | PPT

Thought: The search results don't provide clear, evidence-based treatment recommendations from reputable sources. I should try searching again with a more specific query.

Action: `search_medical_database('evidence-based treatment for the common cold')`

Observation: retrieved 3 results — Manage Common Cold | Common Cold | CDC; The common cold: The need for an effective treatment amid the ...; The common cold in adults: Treatment and prevention - UpToDate

Thought: Reputable sources like the CDC and UpToDate provide guidance on managing common cold symptoms, but the search results don't explicitly state a specific treatment. However, they imply that treatment focuses on symptom management and prevention.

Final answer produced.

Example B — "How to cure acne marks?"

How it works

1. Think about the question
2. Search a medical knowledge base (DuckDuckGo as a PubMed stand-in)
3. Observe the results
4. Answer, grounded in sources

Educational demo only — not medical advice.

Deploy

Healthcare Q&A Agent

ReAct agent · provider: `grog` · live web search

Ask a health question

how to cure acne marks ?

Ask

Answer

Treatments for acne marks include skin needling to stimulate collagen formation, laser treatments like the Er:glass laser for atrophic scars, and subcision for rolling scars. This is general information, not a substitute for professional medical advice.

Sources

- <https://www.mayoclinic.org/diseases-conditions/acne/expert-answers/acne-scars/faq-20058101>
- <https://pmc.ncbi.nlm.nih.gov/articles/PMC5749614/>
- <https://www.nhs.uk/conditions/acne/complications/>

A different topic, same clean behavior, with Mayo/NIH/NHS sources.

Deploy

How it works

1. Think about the question
2. Search a medical knowledge base (DuckDuckGo as a PubMed stand-in)
3. Observe the results
4. Answer, grounded in sources

Educational demo only — not medical advice.

Sources

- <https://www.mayoclinic.org/diseases-conditions/acne/expert-answers/acne-scars/faq-20058101>
- <https://pmc.ncbi.nlm.nih.gov/articles/PMC5749614/>
- <https://www.nhs.uk/conditions/acne/complications/>

Reasoning trace

Steps taken: 2

Thought: To address acne marks, I need to find information on treatments for acne scars and hyperpigmentation.

Action: `search_medical_database('treatments for acne marks and scars')`

Observation: retrieved 3 results — Acne scars: What's the best treatment? - Mayo Clinic; Acne Scarring—Pathogenesis, Evaluation, and Treatment Options; Complications - Acne - NHS

Thought: The search results provide various treatments for acne marks, including skin needling, laser treatments, and subcision.

Final answer produced.

4. A full reasoning trace (text)

The same common-cold run captured from the terminal — the exact reasoning chain, tool calls, and final output:

```
[QUERY] what is treatment for common cold?

[THOUGHT] I need to find evidence-based recommendations for treating the common cold.
[ACTION] search_medical_database('treatment for the common cold')
[OBSERVE] 3 results - weak / non-authoritative sources

[THOUGHT] These results aren't from reputable sources. I should search again
with a more specific query.
[ACTION] search_medical_database('evidence-based treatment for the common cold')
[OBSERVE] 3 results - CDC; review article; UpToDate

[THOUGHT] Reputable sources (CDC, UpToDate) point to symptom management and prevention.
[ANSWER] The common cold is typically managed by alleviating symptoms - OTC
pain/fever medicine, rest, and hydration. This is general information,
not a substitute for professional medical advice.
[SOURCES] cdc.gov/common-cold/treatment, jaci-global.org/..., uptodate.com/...
```

5. Evaluation

5 scenarios in tests/scenarios.json, run by evaluate.py, checking each answer against simple expectations. It runs fully offline (mock LLM + fixtures), so results are deterministic.

#	Scenario	What it checks	Result
1	Type 2 diabetes treatment	mentions metformin, cites sources	PASS

2	First-line hypertension therapy	mentions a first-line drug class	PASS
3	Asthma long-term treatment	mentions corticosteroid	PASS
4	Empty query	guard-rail rejects the input	PASS
5	Unknown topic	falls back to general trusted sources	PASS

Score: 5/5.

6. Design decisions & trade-offs

- **DuckDuckGo instead of PubMed.** Same engineering for free; trade-off is lower medical authority — acceptable given the assignment values engineering over accuracy.
- **Custom ReAct over LangChain.** Smaller, clearer, no dependency churn; trade-off is no built-in memory, which this scope doesn't need.
- **Mock LLM + offline fixtures.** The project and its evaluation run with no key and no network; switching to Groq is one env-var change.
- **Degrade gracefully.** LLM error → retry then fallback; search failure → fixtures; too many steps → safe answer. It never just crashes on the user.

7. How to run it

Try it live (no setup): <https://healthcare-agent-fo48sgeutc25uy2ogrqp6.streamlit.app/>

```

pip3 install -r requirements.txt
cp .env.example .env          # add your free Groq key (console.groq.com/keys)

# Ask a question (Groq + live web search):
python3 -m src.agent --domain healthcare --query "Treatment options for Type 2 diabetes?"

# Or launch the UI:
python3 -m streamlit run app.py

```

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